

ST BRIEUC ARMOR AP, France

WMO: 071200

Lat: 48.5364N

Lon: 2.8532W

Elev: 138

StdP: 99.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -1.9	(c) -0.5	(d) -5.4	(e) 2.4	(f) -0.3	(g) -3.8	(h) 2.8	(i) 0.5	(j) 13.6	(k) 10.1	(l) 12.4	(m) 9.6	(n) 3.1	(o) 130	(p) 0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 8.1	(c) 25.5	(d) 18.9	(e) 23.3	(f) 17.9	(g) 21.9	(h) 17.2	(i) 19.7	(j) 23.8	(k) 18.7	(l) 22.1	(m) 17.9	(n) 20.9	(o) 3.8	(p) 50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.3	21.4	17.3	12.6	20.2	16.7	12.1	19.4	56.6	24.0	53.5	22.2	51.0	21.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.0	9.6	8.6	-4.2	30.4	2.4	2.8	-6.0	32.4	-7.4	34.0	-8.7	35.6	-10.5	37.6
			-5.1	21.7	2.3	1.5	-6.7	22.7	-8.1	23.6	-9.3	24.4	-11.0	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	6.2	6.5	7.8	9.4	12.2	15.0	17.0	17.1	15.2	12.7	9.2	6.8
	DBStd	4.80	3.30	3.00	2.81	2.78	2.60	2.58	2.14	2.20	2.42	2.72	2.65	3.34
	HDD10.0	510	120	103	79	42	8	1	0	0	1	7	44	106
	HDD18.3	2601	375	332	326	267	190	105	52	52	98	175	274	356
	CDD10.0	985	3	4	12	26	77	151	217	219	157	92	21	8
	CDD18.3	34	0	0	0	0	1	5	11	12	5	1	0	0
	CDH23.3	191	0	0	0	2	6	44	55	56	24	3	0	0
	CDH26.7	39	0	0	0	0	1	11	11	13	3	1	0	0
(14) Wind	WSAvg	4.3	5.0	4.9	4.6	4.4	4.0	3.8	3.8	3.6	3.8	4.0	4.4	4.7
(15) Precipitation	PrecAvg	769	79	70	53	61	55	46	46	48	49	78	92	92
	PrecMax	990	161	145	153	170	134	120	137	105	145	167	177	209
	PrecMin	507	10	13	9	14	18	7	13	15	6	13	27	21
	PrecStd	111	42	34	33	36	24	29	27	25	29	34	41	42
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	15.4	18.0	21.8	23.9	28.2	27.7	28.7	26.6	22.0	16.8	14.3
		MCWB	12.1	11.3	12.2	14.3	16.7	20.3	19.9	20.2	18.4	17.1	14.9	12.8
	2%	DB	12.7	12.9	15.0	17.9	20.2	23.8	24.6	24.6	22.9	19.2	15.3	13.5
		MCWB	11.5	10.8	11.4	13.4	15.6	18.2	18.5	19.0	17.4	16.0	13.7	12.2
	5%	DB	11.8	11.8	13.3	15.7	18.3	21.3	22.7	22.8	20.9	17.9	14.2	12.8
		MCWB	10.6	10.2	11.0	11.9	14.5	16.6	17.6	18.1	16.7	15.3	12.7	11.6
	10%	DB	10.9	10.9	12.1	13.8	16.7	19.7	21.2	21.1	19.3	16.7	13.2	11.8
		MCWB	9.7	9.4	10.2	10.7	13.4	15.8	16.9	17.1	15.9	14.7	11.8	10.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	12.2	13.3	15.3	17.9	20.6	20.8	21.3	19.6	17.7	15.5	13.3
		MCDB	13.4	14.0	16.0	19.4	22.1	26.3	26.5	26.3	23.2	20.3	16.3	14.1
	2%	WB	11.8	11.3	12.2	13.5	16.2	18.6	19.2	19.7	18.3	16.7	14.0	12.5
		MCDB	12.6	12.6	14.2	17.1	19.7	23.0	23.0	23.4	21.6	18.6	14.9	13.3
	5%	WB	10.8	10.5	11.3	12.3	14.9	17.2	18.4	18.7	17.3	15.8	13.0	11.8
		MCDB	11.7	11.6	13.0	15.1	17.5	20.5	21.6	21.7	20.0	17.4	14.1	12.7
	10%	WB	9.9	9.6	10.4	11.2	13.9	16.1	17.5	17.8	16.5	15.0	12.0	10.8
		MCDB	10.8	10.7	12.1	13.4	16.2	19.0	20.6	20.3	18.8	16.4	13.1	11.7
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.5	6.4	7.3	7.5	8.0	8.1	8.1	7.7	6.4	5.2	4.9
		MCDBR	5.2	6.4	7.9	9.8	9.8	10.7	10.3	10.4	9.7	7.3	5.8	5.1
		MCWBR	4.8	4.9	5.3	5.9	5.9	6.0	5.3	5.4	5.1	4.6	4.8	4.7
	5% WB	MCDBR	5.1	6.0	7.2	8.9	9.2	9.9	9.3	9.2	8.9	6.7	5.6	5.2
		MCWBR	4.9	5.1	5.2	5.8	6.0	6.0	5.4	5.3	5.3	4.8	5.0	5.1
(40) Clear-Sky Solar Irradiance	taub		0.326	0.342	0.371	0.384	0.389	0.384	0.382	0.372	0.359	0.353	0.331	0.320
	taud		2.417	2.389	2.314	2.294	2.323	2.364	2.373	2.422	2.440	2.452	2.441	2.390
	Ebn at Noon		734	806	838	864	873	877	873	868	844	785	721	685
	Edh at Noon		67	85	107	121	122	118	115	105	94	79	65	61
(44) All-Sky Solar Radiation	RadAvg		0.96	1.74	2.81	4.32	5.12	5.63	5.37	4.56	3.61	2.10	1.17	0.81
	RadStd		0.08	0.20	0.29	0.42	0.42	0.62	0.33	0.37	0.27	0.17	0.12	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page